

Temperature Analysis Report

The temperature profiles obtained from four sensors, namely Top Left (Temp1), Room Temperature (Temp2), Middle Right End (Temp3), and Middle (Temp4), provide valuable insights into the thermal dynamics of a controlled environment, presumably inside a large box. The statistical summaries and correlation matrix offer a detailed understanding of temperature variations within and outside the box. The experimental observation was done for over 330 hours or 13.75 days which is roughly 2 weeks. This was tested using new PID values, by varying the damping factor to achieve a desirable output. The overall objective was to test the new PID values and see how efficient the system's maintaining temperature inside with respect to fluctuations in room temperature.

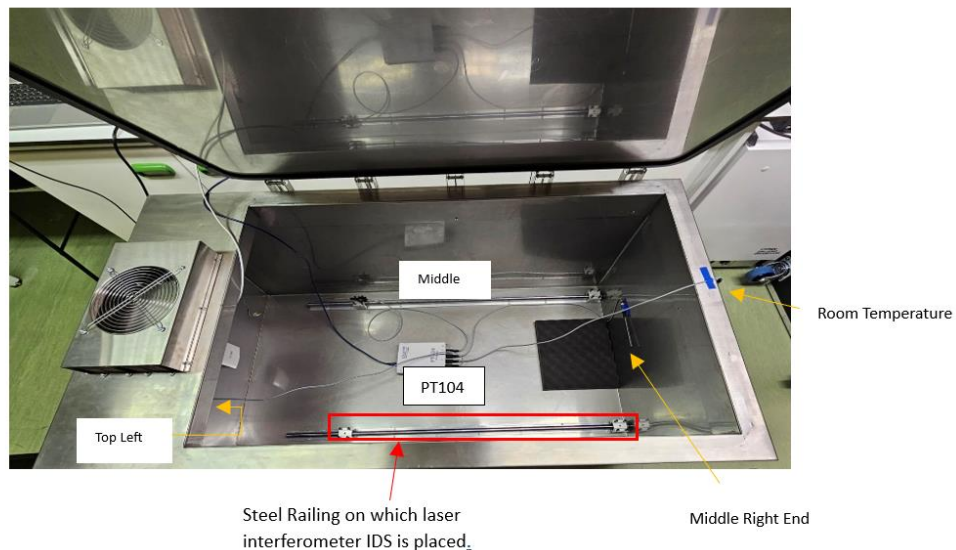


Fig 1. Shows the position of all the temperature probes.

❖ Temperature Statistics:

- **Temp1 (Top Left):**
Mean: 20.82°C, Mode: 20.833, Standard Deviation: 0.15°C
- **Temp2 (Room Temperature):**
Mean: 19.67°C, Mode: 20.008°C, Standard Deviation: 0.49°C
- **Temp3 (Middle Right End):**
Mean: 20.55°C, Mode: 20.477°C, Standard Deviation: 0.22°C
- **Temp4 (Middle):**
Mean: 20.58°C, Mode: 20.532°C, Standard Deviation: 0.16°C

The Fig 2. reveals the trend of temperature variations, one of the major fluctuations we can see in the range 50-60h, the box temperature control (Top Left) however maintains a constant, but a slight variation in change of about 0.15°C at 50th hour of Top Left with respect to the middle of the box. Let's discuss in brief below.

The analysis of temperature data obtained from the four sensors—Top Left (Temp1), Room Temperature (Temp2), Middle Right End (Temp3), and Middle (Temp4)—revealed insightful distribution characteristics. Figure 3 showcases histograms providing a visual representation of the frequency distribution for each temperature profile. These histograms include kernel density estimates (KDE) to offer a smoother depiction of the distribution patterns. A notable observation is the distinct distribution of room temperature fluctuations, evident in Temp2, which displays more bins compared to other temperature profiles. Temp1, representing the top-left sensor, demonstrates a single bin,

indicating a relatively constant temperature—a desirable characteristic for the intended purpose. However, a subtle variation is observed in Temp3 and Temp4, each showing two bins in their histograms. This visual analysis contributes to a understanding of temperature variations across different sensors, emphasizing the effectiveness of maintaining a consistent temperature in the top-left region while highlighting slight fluctuations in other areas.

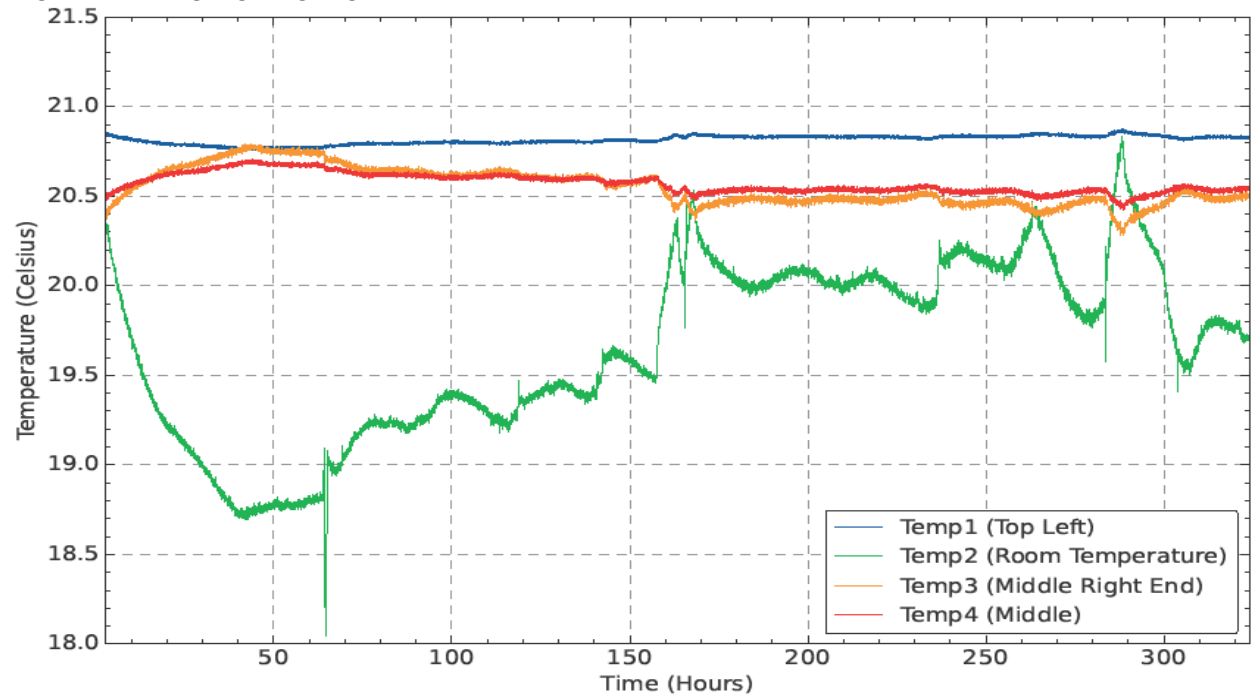


Fig 2. Shows the temperature data collected using 4 temperature probes over 2 weeks.

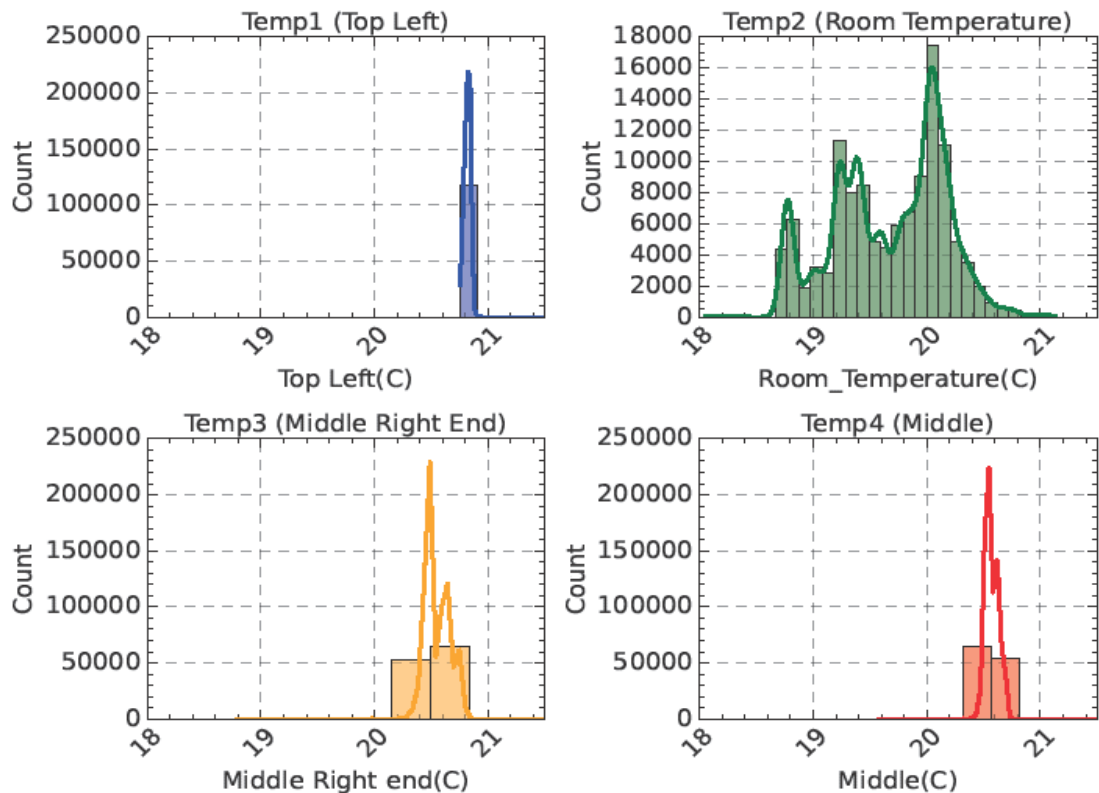


Fig 3. Shows the variation in data in form of histograms.

For better understanding, I combined each of the different histograms (fig 4) into a single representation in the next section. This combined perspective offers a more understandable understanding of how room temperature varies. Temp1 displays the intended consistent temperature as expected. Inside the box, there is space for development, especially when it comes to maximizing air circulation. This enhancement is necessary to lessen the minor temperature swings that Temp3 and Temp4 show, finally bringing them closer to the constant temperature that Temp1 displays. By taking care of this, the box's temperature environment can be made more consistent and under control.

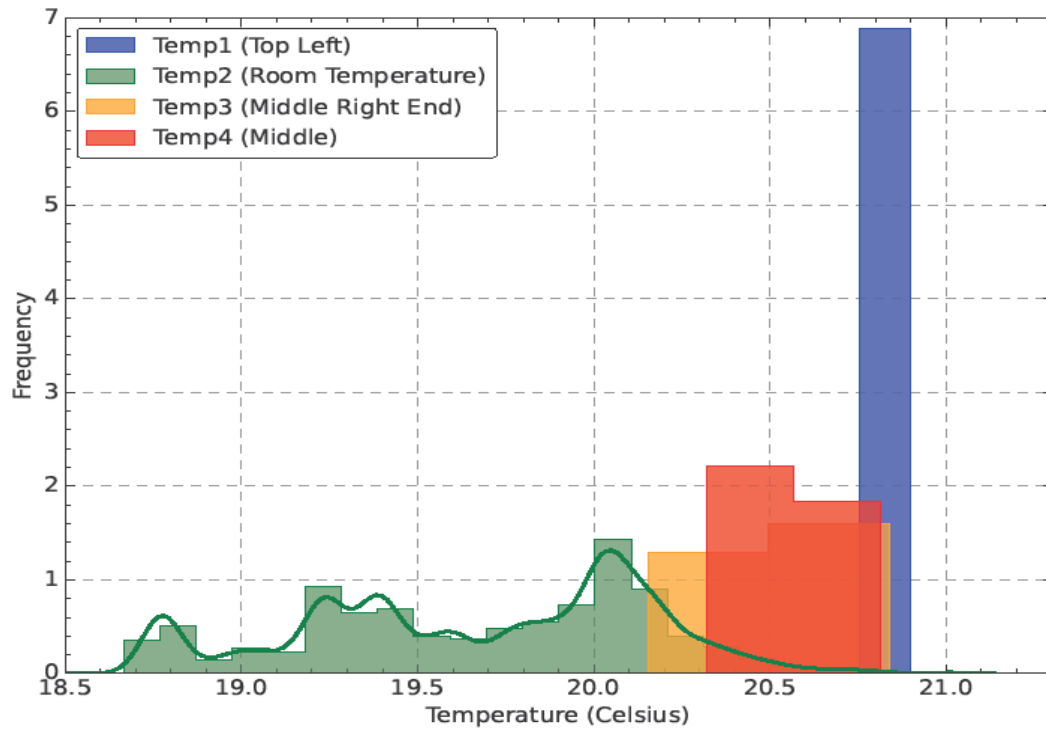


Fig 4. Shows the combined variation in data in form of histograms.

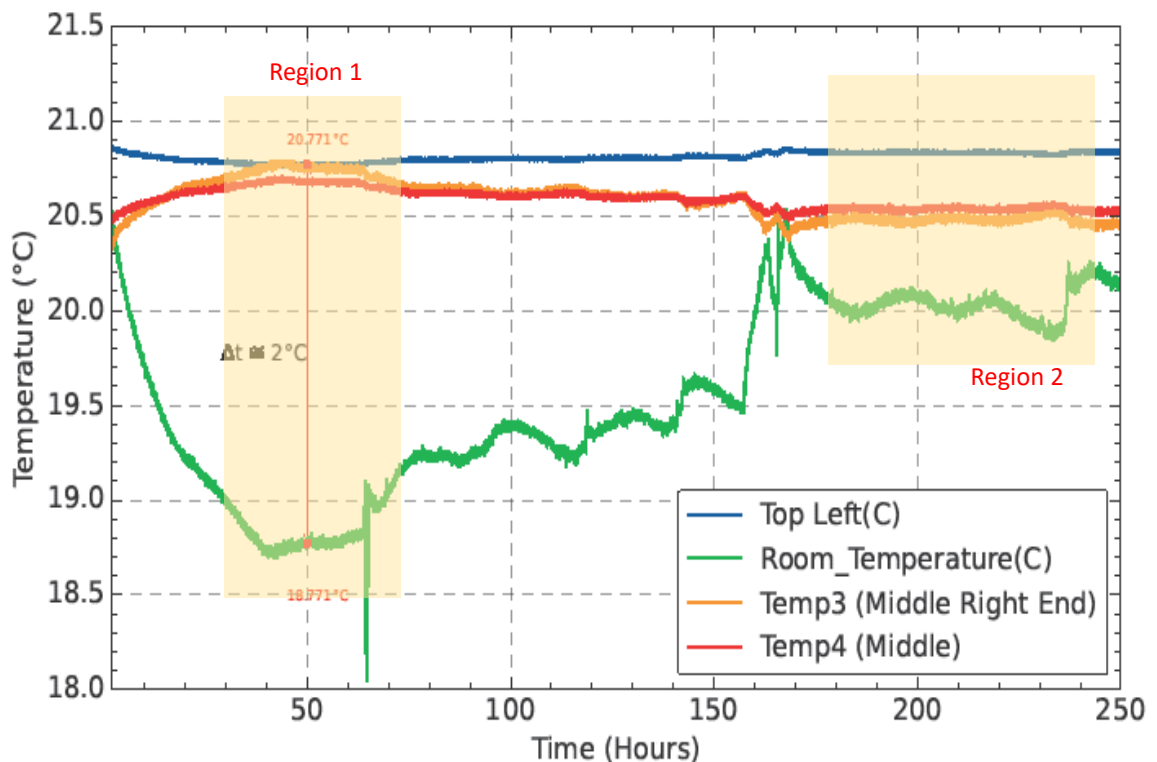


Fig 5. Region of focus: Region 1 and Region 2

Now let's look closely at two particular region of interests – in region 1 we can see a big spike of about 2 degrees Celsius of Room Temperature with respect to temp1, however the internal temperature is constant for 40hrs, and 2nd region we see due to variation in room temperature, there is variation in inside of the box, however in this case the temp1 constant(expected), temp3 and temp 4 correlated and constant.

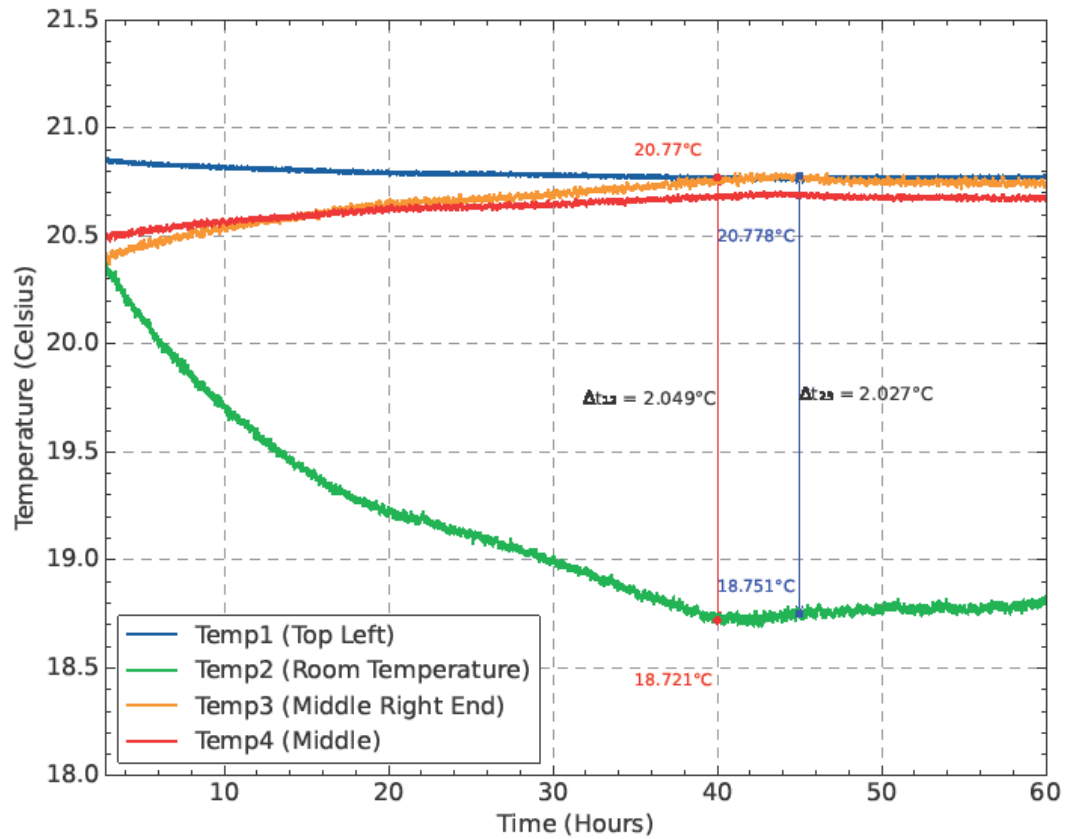


Fig 6. Region 1 analysis

Next, evaluated the time variations in Figure 6, which shows that the box's inside temperature remained constant over the observation period. The temperature differential of the interior environment (20.77 degrees Celsius) and the room (18.721 degrees Celsius) is roughly 2.049 degrees Celsius. This data is consistent with our hypothesis, confirming that the box can maintain a steady temperature even when there is an outside temperature surge of two degrees Celsius. It's important to keep in mind that although if the box can handle an external temperature change of two degrees, larger outside temperature variations may cause the reaction time to increase. Since it would probably take longer to reach and sustain, more research is needed to see how the box reacts to greater temperature fluctuations.

Similarly, in Region 2, we conducted an analysis where we calculated the following temperature differences:

Δt_{12} (difference between Temp1 and Temp2): 0.928°C

Δt_{234} (difference between the average of Temp3 and Temp4 and Temp2): 0.65°C

Δt_{134} (difference between Temp1 and the average of Temp3 and Temp4): 0.278°C.

These calculations provide insights into the temperature variations between different sensors, highlighting specific differences within Region 2.

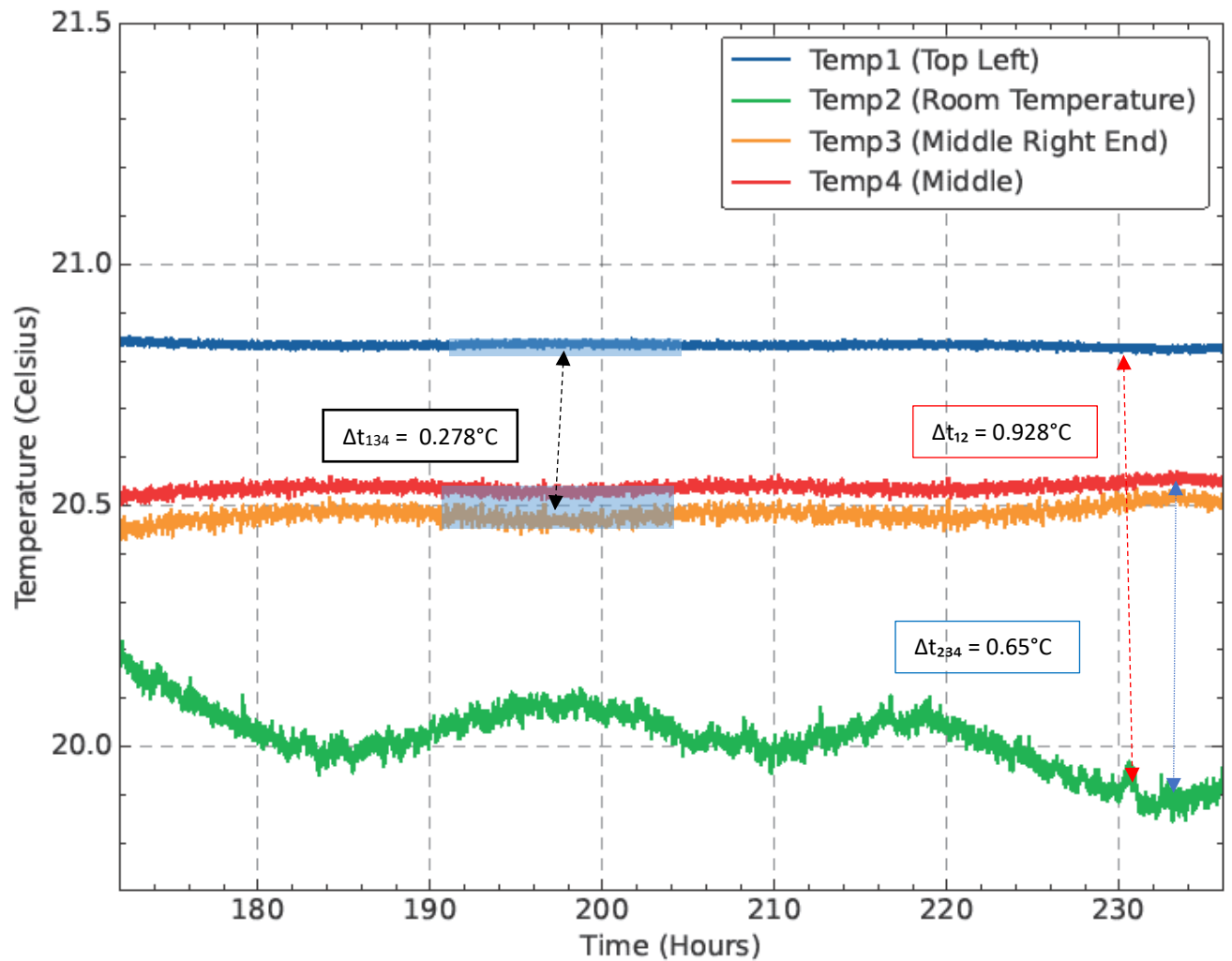


Fig 7. Region 2 analysis

Next, Performed an overall statistical analysis on the overall data of the room temperature vs the box temp1.

Mean Temperature Difference: -1.1497183283864267

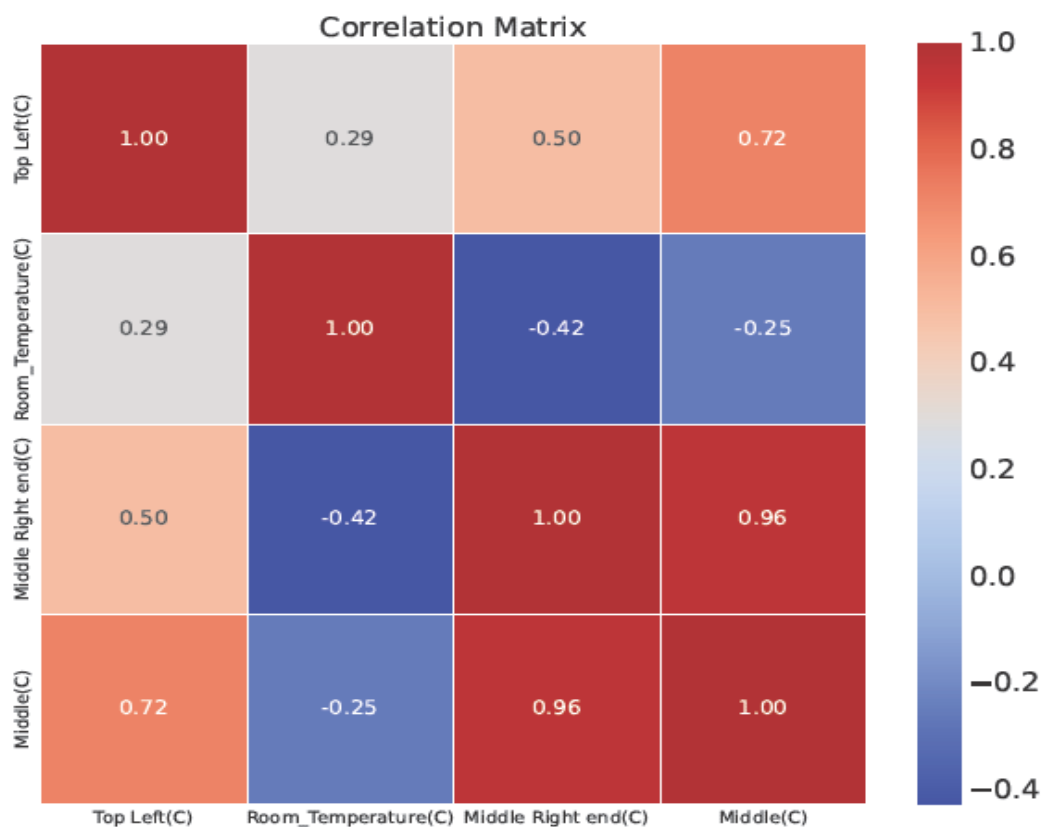
Standard Deviation of Temperature Difference: 0.47327475586196827

In the conducted analysis, the mean temperature difference between the observed temperatures was found to be -1.1497°C . This negative value suggests an overall decrease in temperature. Additionally, the standard deviation of the temperature difference was calculated to be 0.4733°C , indicating a certain degree of variability or dispersion in the temperature differences.

These statistical measures provide insights of the central tendency and spread of temperature differences within the dataset. The negative mean implies a consistent cooling trend, while the standard deviation quantifies the extent to which individual temperature variations deviate from this average change. This information is instrumental in characterizing and understanding the thermal dynamics captured by the collected data.

❖ Correlation Matrix:

- **Temp1 vs. Temp2 (0.29):**
Weak positive correlation, indicating some influence of room temperature changes on the Top Left area.
- **Temp1 vs. Temp3 (0.50):**
Moderate positive correlation, suggesting a connected thermal response between Top Left and Middle Right End.
- **Temp1 vs. Temp4 (0.72):**
Strong positive correlation, indicating a consistent temperature pattern between Top Left and Middle.
- **Temp2 vs. Temp3 (-0.42):**
Moderate negative correlation, implying differential responses to external changes in Room Temperature and Middle Right End.
- **Temp3 vs. Temp4 (0.96):**
Very strong positive correlation, revealing a synchronized thermal behavior between the Middle Right End and Middle.



❖ Analysis:

The results suggest that the box effectively maintains a constant temperature, especially in the region close to the Top Left probe. However, variations are observed in the Middle and Middle Right End areas. The correlation matrix indicates intricate relationships between the temperature profiles, with a significant influence of external changes on certain internal regions.

❖ Interpretation:

- The Top Left probe, strategically placed for control, shows stable and consistent temperatures.
- Variations in the Middle and Middle Right End areas suggest potential challenges in achieving rapid temperature stabilization.
- The strong positive correlation between Temp3 and Temp4 indicates a synchronized response in maintaining a constant temperature in these two areas.

❖ Conclusion:

While the box demonstrates effective temperature control, further optimization may be considered for achieving quicker stability in all internal areas. Understanding the correlations helps identify areas with shared thermal behaviour and informs potential adjustments to enhance overall performance.

In summary, the analysis provides a comprehensive overview of the temperature dynamics within the controlled environment, offering insights for fine-tuning the temperature control system.